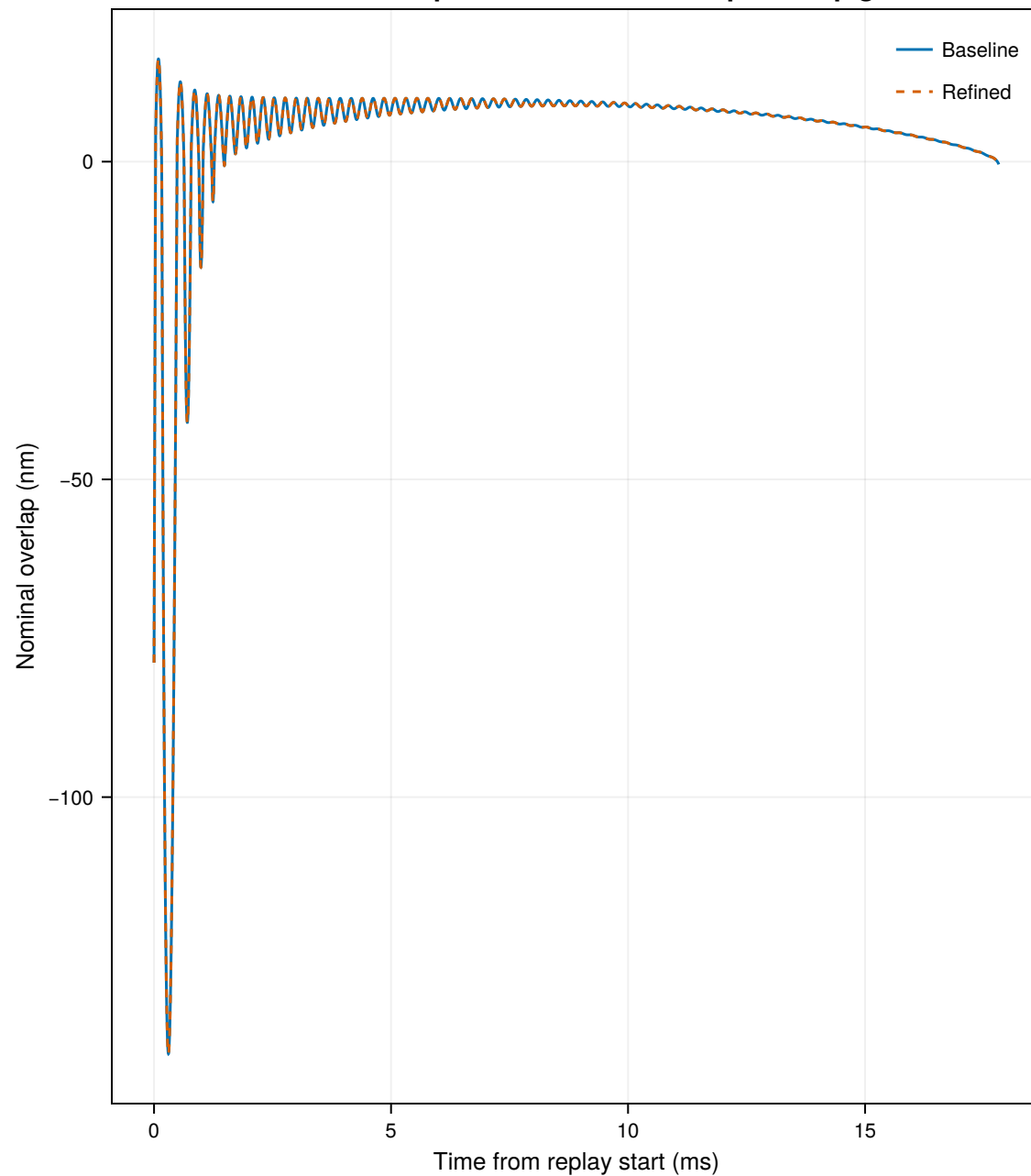


Full-sequence solver-refinement comparison
Side $r = 1$; all rebounds included; same initial state; tolerances $/ 5$ and $dt_{\max} / 2$

Contact overlap on the baseline accepted-step grid



Refined minus baseline at accepted-step times

